Mean Absolute Deviation (MAD)

It is the average of the absolute deviations from the arithmetic mean.

- **For ungrouped data:**

$$\text{MAD} = \frac{\sum |x_i - \bar{x}|}{n}$$

- **For grouped data:**

$$\text{MAD} = \frac{\sum f_i |x_i - \bar{x}|}{N} \quad \text{where } N = \sum f_i$$

Example 1

Find the quartile deviation and the mean absolute deviation for the data: \\$3, 6, 9, 10, 7, 12, 13, 15, 6, 5, 13$

Solution

Sorted data: $3, 5, 6, 6, 7, 9, 10, 12, 13, 13, 15 \quad (n = 11)$ \\ Recall from earlier calculations that $\bar{x} = 9$.

- **Quartiles:**

$$\begin{aligned} \text{Position of } Q_1 &= \frac{11+1}{4} = \text{3rd item} \implies Q_1 = 6 \\ \text{Position of } Q_3 &= \frac{3(11+1)}{4} = \text{9th item} \implies Q_3 = 13 \\ \text{SIQR} &= \frac{13-6}{2} = 3.5 \end{aligned}$$

- **Mean Absolute Deviation (MAD):**

$$\begin{aligned} \sum |x_i - \bar{x}| &= |3-9| + |5-9| + 2|6-9| + |7-9| + |9-9| + |10-9| + |12-9| + 2|13-9| + |15- \\ &= 6+4+6+2+0+1+3+8+6 = 36 \\ \text{MAD} &= \frac{36}{11} \approx 3.27 \end{aligned}$$

Variance and Standard Deviation

Ignoring the negative sign in order to compute MAD is not the only option we have to deal with deviations. We can square the deviations and then average them. The average of the squared deviations from the mean is called the **variance**, denoted by s^2 (sample) or σ^2 (population):

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n}$$

A little algebraic simplification of this formula gives:

$$s^2 = \frac{\sum x_i^2}{n} - \bar{x}^2$$

For grouped data, where N is the sum of frequencies ($\sum f_i$):

$$s^2 = \frac{\sum f_i x_i^2}{N} - \bar{x}^2$$

To reverse the squaring on the units, we find the square root of the variance. The **Standard Deviation**, denoted by s, is the square root of the variance:

$$s = \sqrt{\frac{\sum x_i^2}{n} - \bar{x}^2}$$

Example 1

Find the variance and standard deviation for the data: $3, 6, 9, 10, 7, 12, 13, 15, 6, 5, 13$.

Solution

We know $n = 11$ and $\bar{x} = 9$.

$$\sum x_i^2 = 3^2 + 5^2 + 6^2 + 6^2 + 7^2 + 9^2 + 10^2 + 12^2 + 13^2 + 13^2 + 15^2 = 1042$$

$$\text{Variance } (s^2) = \frac{1042}{11} - 9^2 = 94.73 - 81 = 13.73$$

$$\text{Standard Deviation } (s) = \sqrt{13.73} \approx 3.71$$

Example 2

Find the standard deviation of the data: $2, 4, 8, 7, 9, 4, 6, 10, 8,$ and 5.

Solution

Sorted Array: $2, 4, 4, 5, 6, 7, 8, 8, 9, 10 \quad (n = 10)$

$$\bar{x} = \frac{63}{10} = 6.3$$

$$\sum x_i^2 = 2^2 + 4^2 + 4^2 + 5^2 + 6^2 + 7^2 + 8^2 + 8^2 + 9^2 + 10^2 = 455$$

$$\text{Variance } (s^2) = \frac{455}{10} - (6.3)^2 = 45.5 - 39.69 = 5.81$$

$$\text{Standard Deviation } (s) = \sqrt{5.81} \approx 2.41$$

Example 3

Estimate the mean and standard deviation for the frequency table below:

Class	5–9	10–14	15–19	20–24	25–29	30–34	35–39
Frequency (f)	5	12	32	40	16	9	6

Midpoints (x)	Frequency (f)	xf	x^2	x^2f
7	5	35	49	245
12	12	144	144	1728
17	32	544	289	9248
22	40	880	484	19360
27	16	432	729	11664
32	9	288	1024	9216
37	6	222	1369	8214
Total	$N = 120$	$\sum xf = 2545$		$\sum x^2 f = 59675$

Solution

$$\text{Mean } (\bar{x}) = \frac{2545}{120} \approx 21.21$$

$$\text{Variance } (s^2) = \frac{59675}{120} - (21.2083)^2 = 497.2917 - 449.7921 = 47.4996$$

$$\text{Standard Deviation } (s) = \sqrt{47.4996} \approx 6.89$$

Exercise Solutions

Question 1

Find the quartile deviation (SIQR), mean absolute deviation (MAD), and standard deviation (s) of:

a) $9, 3, 4, 2, 9, 5, 8, 4, 7, 4 \implies$ **Sorted:** $2, 3, 4, 4, 4, 5, 7, 8, 9, 9 \quad (n = 10)$

- $\bar{x} = \frac{55}{10} = 5.5$
- $Q_1 =$ value at position $\frac{10+1}{4} = 2.75 \implies 3 + 0.75(4-3) = 3.75$
- $Q_3 =$ value at position $\frac{3(11)}{4} = 8.25 \implies 8 + 0.25(9-8) = 8.25$
- Quartile Deviation (SIQR) $= \frac{8.25-3.75}{2} = 2.25$
- MAD $= \frac{|2-5.5|+|3-5.5|+3|4-5.5|+|5-5.5|+|7-5.5|+|8-5.5|+2|9-5.5|}{10} = \frac{22}{10} = 2.2$
- $\sum x^2 = 368 \implies s = \sqrt{\frac{368}{10} - 5.5^2} = \sqrt{36.8 - 30.25} = \sqrt{6.55} \approx 2.56$

b) $1, 2, 2, 3, 4, 4, 5, 5, 5, 5, 7, 8, 8, 9 \quad (n = 14)$

- $\bar{x} = \frac{68}{14} \approx 4.86$
- $Q_1 =$ position $\frac{15}{4} = 3.75 \implies 2 + 0.75(3-2) = 2.75$
- $Q_3 =$ position $\frac{45}{4} = 11.25 \implies 7 + 0.25(8-7) = 7.25$
- Quartile Deviation (SIQR) $= \frac{7.25-2.75}{2} = 2.25$
- MAD $= \frac{\sum |x_i - 4.857|}{14} = \frac{26.286}{14} \approx 1.88$
- $\sum x^2 = 408 \implies s = \sqrt{\frac{408}{14} - (4.857)^2} = \sqrt{29.143 - 23.590} = \sqrt{5.553} \approx 2.36$

Question 2

No. of goals (x)	1	2	3	4	5
No. of matches (f)	2	5	6	3	4

Solution: $N = 20, \sum fx = 62 \implies \bar{x} = \frac{62}{20} = 3.1$ goals.

- **Quartiles:** Cumulative frequencies are $2, 7, 13, 16, 20$. Q_1 pos $= \frac{20}{4} = 5 \implies Q_1 = 2$. Q_3 pos $= \frac{3(20)}{4} = 15 \implies Q_3 = 4$. SIQR $= \frac{4-2}{2} = 1.0$.
- MAD $= \frac{2|1-3.1|+5|2-3.1|+6|3-3.1|+3|4-3.1|+4|5-3.1|}{20} = \frac{4.2+5.5+0.6+2.7+7.6}{20} = \frac{20.6}{20} = 1.03$
- $\sum fx^2 = 222 \implies s = \sqrt{\frac{222}{20} - 3.1^2} = \sqrt{11.1 - 9.61} = \sqrt{1.49} \approx 1.22$

Class	8–12	13–17	18–22	23–27	28–32	33–37
Freq (f)	3	10	12	9	5	1

Question 3

Solution Matrix: $N = 40$. Midpoints (x): $10, 15, 20, 25, 30, 35$. $\sum fx = 805 \implies \bar{x} = 20.125$. $\sum fx^2 = 17625$.

- **Quartiles (Grouped LCB method):** Class width $i = 5$. Q_1 class (at 10th item) $= 12.5 - 17.5 \implies Q_1 = 12.5 + \left(\frac{10-3}{10}\right) 5 = 16.0$ Q_3 class (at 30th item) $= 22.5 - 27.5 \implies Q_3 = 22.5 + \left(\frac{30-25}{9}\right) 5 \approx 25.28$ SIQR $= \frac{25.28-16.0}{2} = 4.64$
- MAD $= \frac{\sum f|x-20.125|}{40} = \frac{203.75}{40} \approx 5.09$
- $s = \sqrt{\frac{17625}{40} - (20.125)^2} = \sqrt{440.625 - 405.016} = \sqrt{35.609} \approx 5.97$

Question 4

Height (cm)	140–144	144–148	148–152	152–156	156–160	160–164
Students (f)	4	5	8	7	5	1

Solution Matrix: $N = 30$. Midpoints (x): $142, 146, 150, 154, 158, 162$. $\sum fx = 4504 \implies \bar{x} = 150.13$ cm. $\sum fx^2 = 677104$.

- **Quartiles:** Width $i = 4$. Q_1 class (pos 7.5) $= 144 - 148 \implies Q_1 = 144 + \left(\frac{7.5-4}{5}\right) 4 = 146.8$ cm Q_3 class (pos 22.5) $= 152 - 156 \implies Q_3 = 152 + \left(\frac{22.5-17}{7}\right) 4 \approx 155.14$ cm SIQR $= \frac{155.14-146.8}{2} = 4.17$ cm
- MAD $= \frac{\sum f|x-150.13|}{30} = \frac{133.08}{30} \approx 4.44$ cm
- $s = \sqrt{\frac{677104}{30} - (150.133)^3} = \sqrt{22570.133 - 22540.018} = \sqrt{30.115} \approx 5.49$ cm

Question 5

Distance (d)	0–5	5–10	10–15	15–20	20–25	25–30
People (f)	15	28	40	35	20	12

Solution Matrix: $N = 150$. Midpoints (x): $2.5, 7.5, 12.5, 17.5, 22.5, 27.5$. $\sum fx = 2102.5 \implies \bar{x} = 14.02$ km. $\sum fx^2 = 36562.5$.

- **Quartiles:** Width $i = 5$. Use code with caution.Q_1 class (pos 37.5) $= 5 - 10 \implies Q_1 = 5 + \left(\frac{37.5-15}{28}\right) 5 \approx 9.02$ kmQ_3 class (pos 112.5) $= 15 - 20 \implies Q_3 = 15 + \left(\frac{112.5-83}{35}\right) 5 \approx 19.21$ kmSIQR $= \frac{19.21-9.02}{2} = 5.10$ km
- $s = \sqrt{\frac{36562.5}{150} - (14.0167)^2} = \sqrt{243.75 - 196.468} = \sqrt{47.282} \approx 6.88$ km

Properties of Measures of Spread

a) **Change of Origin:** They are not affected by a change of origin. Adding or subtracting a constant from each and every observation in a dataset does not affect any measure of spread:
$$\text{Spread}(X \pm c) = \text{Spread}(X)$$

b) **Change of Scale:** They are affected by a change of scale. Multiplying each and every observation in a dataset by a constant value scales up all standard measures of spread by the absolute value of that constant, except for variance, which scales up by the square of that constant:

Assumed Mean and the Coding Formula

If the observations are large such that the natural computation of totals becomes tedious, we can take one of the observations as a working or assumed mean. Let A be any guessed or assumed arithmetic mean and let $d_i = x_i - A$ be the deviations of x_i from A. Then the arithmetic mean ($\bar{x}$) and variance (s^2) are respectively given by:

$$\bar{x} = A + \frac{\sum f_i d_i}{N} \quad \text{and} \quad s^2 = \frac{\sum f_i d_i^2}{N} - \left(\frac{\sum f_i d_i}{N}\right)^2$$

Where A is the assumed mean, which is generally taken as the midpoint of the middle class or the class where the frequency is largest, and $N = \sum f_i$.

Remark: In most cases, the deviations (d_i) of x_i from A are multiples of the class interval size (i). That is:

$$t_i = \frac{x_i - A}{i}$$

In these cases, we can use the coded variable t rather than d in the computation. The equations reduce to:

$$\bar{x} = A + \left(\frac{\sum f_i t_i}{N}\right) \times i \quad \text{and} \quad s^2 = i^2 \left[\frac{\sum f_i t_i^2}{N} - \left(\frac{\sum f_i t_i}{N}\right)^2\right]$$

These reduced formulations are referred to as the **coding formulas**.

Example

Using the coding formula, find the mean and standard deviation of the following data:

Class	6340–6349	6350–6359	6360–6369	6370–6379	6380–6389
Frequency (f)	2	3	7	5	3

Solution Matrix Setup

Let the assumed mean be the midpoint of the middle class: $A = 6364.5$. \The class interval size is $i = 10$.

Class	Midpoints (x)	Frequency (f)	Coded Variable (t)	ft	ft^2
6340 – 6349	6344.5	2	-2	-4	8
6350 – 6359	6354.5	3	-1	-3	3
6360 – 6369	6364.5	7	0	0	0
6370 – 6379	6374.5	5	1	5	5
6380 – 6389	6384.5	3	2	6	12
Total	–	$N = 20$	–	$\sum ft = 4$	$\sum ft^2 = 28$

1. Calculation of the Arithmetic Mean ($\bar{x}$)

$$\begin{aligned}\bar{x} &= A + \left(\frac{\sum f_i t_i}{N}\right) \times i \\ &= 6364.5 + \left(\frac{4}{20}\right) \times 10 \\ &= 6364.5 + (0.2 \times 10) \\ &= 6364.5 + 2 \\ &= 6366.5\end{aligned}$$

2. Calculation of the Variance (s^2) and Standard Deviation (s)

$$\begin{aligned}s^2 &= i^2 \left[\frac{\sum f_i t_i^2}{N} - \left(\frac{\sum f_i t_i}{N}\right)^2\right] \\ &= 10^2 \left[\frac{28}{20} - \left(\frac{4}{20}\right)^2\right] \\ &= 100\left[1.4 - (0.2)^2\right] \\ &= 100\left[1.4 - 0.04\right] \\ &= 100 \times 1.36 = 136\end{aligned}$$

Taking the square root to find the standard deviation (s):

$$s = \sqrt{136} \approx 11.6619$$

Measures of Relative Dispersion

These measures are used in comparing spreads of two or more sets of observations.

These measures are independent of the units of measurement. These are a sort of ratio and are called coefficients.

The smaller the coefficient the lower the spread and vice versa. Suppose that the two distributions to be compared are expressed in the same units and their means are equal or nearly equal.

Then their variability can be compared directly by using their standard deviations. However, if their means are widely different or if they are expressed in different units of measurement, we can not use the standard deviations as such for comparing their variability. We have to use the relative measures of dispersion in such situations.

Examples of these Measures of relative dispersion includes; Coefficient of quartile deviation, Coefficient of mean deviation and the Coefficient of variation

Coefficient of Quartile Deviation

The **Coefficient of Quartile Deviation** of x, denoted as CQD(x), is a relative measure of spread based on the lower (Q_1) and upper (Q_3) quartiles:

$$\text{CQD}(x) = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

Coefficient of Mean Deviation

The **Coefficient of Mean Deviation** of x, denoted as CMD(x), expresses the Mean Absolute Deviation (MAD) relative to either the mean ($\bar{x}$) or the median ($\tilde{x}$):

$$\text{CMD}(x) = \frac{\text{MAD}}{\bar{x}} \quad \text{or} \quad \text{CMD}(x) = \frac{\text{MAD}}{\tilde{x}}$$

Coefficient of Variation

The **Coefficient of Variation** is the percentage ratio of the standard deviation (s) to the arithmetic mean ($\bar{x}$). It is an essential relative metric used to compare variance between datasets with different units or vastly different scales:

$$\text{C.V}(x) = \left(\frac{s}{\bar{x}}\right) \times 100\%$$

Where $\bar{x}$ is the mean and s is the standard deviation of x.

Note: Standard deviation is an *absolute* measure of dispersion, whereas the Coefficient of Variation is a *relative* measure of dispersion.

Worked Examples

Example 1

Consider the distribution of the yields (per plot) of two groundnut varieties.

- **First Variety:** Mean $\bar{x}_1 = 82$ kg, Standard Deviation $s_1 = 16$ kg.
- **Second Variety:** Mean $\bar{x}_2 = 55$ kg, Standard Deviation $s_2 = 8$ kg.

Solution

Calculating the Coefficient of Variation for both varieties:

$$\text{C.V}_{\text{Variety 1}} = \left(\frac{16}{82}\right) \times 100\% \approx 19.51\%$$
$$\text{C.V}_{\text{Variety 2}} = \left(\frac{8}{55}\right) \times 100\% \approx 14.55\%$$

It is apparent that the relative variability in the second variety (14.55%) is less compared to that in the first variety (19.51%). However, if interpreting purely in terms of absolute standard deviation (16,kg vs 8,kg), the raw scale comparison could obscure this relative consistency.